

Surface Wipe Down SOP (1x Surface)

One episode is one full wipe down of a table surface. It runs from a dirty table holding items in the layout to the same table clean, dry, and holding the same items back in the same layout, with the towel back in the supply corner and the used wipe in the bin.

The two grippers have fixed roles and never swap. The **left gripper is the item gripper**: it does nothing but carry items off the surface and back onto it. The **right gripper is the tool gripper**: it handles the spray bottle, the wipe, and the towel. The right gripper never moves an item.

There are two cloths and they are never mixed up. The **wipe** takes the dirt and the liquid off the wet surface and goes straight into the bin after that one pass. The **towel** dries the surface afterwards and goes back to the supply corner. The wipe is never used to dry and the towel is never used on a wet surface.

The spray in Step 2 is the one place both grippers work at once. The right gripper holds the bottle and aims it, and the left gripper presses the trigger. That trigger press is the only time the left gripper touches a tool. It never carries the bottle, and it never touches either cloth.

The order never changes: **clear the items to the staging strip, spray the wipe zone, run the wet pattern with the wipe and bin it, run the dry pattern with the towel, return the items to the layout**. A phase is finished and checked before the next one begins, and a finished phase is never revisited.

Never pull the trigger while an item is still standing in the wipe zone. Never drag or slide an item: every item is lifted clear, carried, and set down. Never carry two items in one trip. Never set a cloth down on the wiped surface. Never put the used wipe back in the supply corner. Never return an item to a surface that is still wet.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera**: 900 mm.
- **cell_type**: bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera is centered on the table and its frame includes the front and back table edges, the whole wipe zone, the staging strip, the supply corner, and the mouth of the bin.
3. The dirt inside the wipe zone is visible on camera before the wet pattern.

4. Both arms are at home with grippers open.
5. The table holds nothing but the layout items and the supplies.
6. The left arm reaches every item in the wipe zone and the whole staging strip without extending to a joint limit. The right arm reaches the supply corner, every corner of the wipe zone, and the mouth of the bin the same way. Narrow the wipe zone until both are true.
7. With the right gripper holding the bottle over any part of the wipe zone, the left gripper reaches the trigger from the left without extending to a joint limit and without the two arms touching. Narrow the wipe zone until this is true too.
8. Arm paths to the staging strip on the left and to the supply corner and the bin on the right are unobstructed, and each arm can cross the table midline without collision.

Materials checklist

1. Three to six items stand on the surface in the layout, each in its home position, none touching another item.
2. Every item stands upright unaided, is empty and dry so nothing spills when it is lifted, and can be lifted by one gripper.
3. The surface is visibly dirty in at least one place inside the wipe zone.
4. The spray bottle stands upright in the supply corner, nozzle open, holding enough liquid for at least six trigger pulls.
5. One wipe lies flat in the supply corner, dry, spread out rather than balled up, and clear of the bottle.
6. One towel lies flat in the supply corner, dry, spread out rather than balled up, and clear of the bottle and of the wipe.
7. The wipe and the towel are easy to tell apart on camera by colour or by size, and each one has its own spot in the supply corner.
8. Both cloths are big enough that the body covers a broad strip of the table when the cloth is pinched at the middle and lowered, and light enough for one gripper to drag across the surface.
9. The bin stands beside the supply corner on the right, open at the top, empty or nearly empty, and steady enough that it does not tip when the wipe drops in.
10. The staging strip is empty.

Workspace layout

- **Wipe zone:** the middle of the table, the only area that gets sprayed and wiped. It stops a hand width in from each table edge, and it is no wider than both arms reach comfortably.
- **Layout home positions:** where the items stand inside the wipe zone at episode start, and where they must stand again at episode end. Nothing is marked or taped out.

- **Staging strip:** a clear strip to the left of the wipe zone, outside it, holding the items while the surface is cleaned. Worked by the left gripper only.
- **Supply corner:** the area to the right of the wipe zone holding the spray bottle, the wipe, and the towel. Worked by the right gripper only.
- **Bin:** the open container beside the supply corner, on the right. It takes the used wipe and nothing else. Worked by the right gripper only.
- **Arm home:** the resting pose for each arm, gripper open and clear of the table.

Vocabulary

- **Arm assignment:** fixed by object, not by where it lies. The **left gripper** moves items and works the wipe zone and the staging strip. The **right gripper** handles the spray bottle, the wipe, and the towel, and works the supply corner and the bin. The one exception is the trigger press in Step 2, which the left gripper does while the right gripper holds the bottle. No object is ever handed from one gripper to the other.
- **Trigger press:** the left gripper closes on the trigger of the bottle the right gripper is holding, squeezes to fire a burst, and opens again. The left gripper never takes any of the bottle weight and never carries the bottle.
- **Idle arm:** the arm not named in a step. It waits at its home, clear of the table, for the whole of that step. It never steadies an object, holds a lid, or assists the working arm. Step 2 names both grippers, so it has no idle arm.
- **Item:** one object standing on the surface at episode start, to be cleared and returned.
- **Home position:** the spot and orientation an item occupied at episode start, judged by eye against the table edges and the neighbouring items.
- **Layout:** all the items in their home positions together.
- **Layout restored:** every item stands upright in its own home position, in the orientation it started in, touching no other item and none left on the staging strip.
- **Wipe:** the cloth used once and then thrown away. It runs the wet pattern, lifting the dirt and the sprayed liquid off the surface, and it goes into the bin as soon as that pattern is finished. It is never used to dry, never set back in the supply corner, and never picked up again.
- **Towel:** the drying cloth. It is dry at episode start, it runs the dry pattern over the damp surface after the wipe is binned, and it goes back to its spot in the supply corner at the end. It never touches a surface that still carries standing liquid.
- **Middle pinch:** the grip used on the wipe and on the towel. The gripper closes on the cloth at its middle, not at an edge or a corner, so the cloth hangs even on both sides of the gripper and its body settles flat on the table. A cloth held by a corner, balled up, or dragged edge on is not a middle pinch.
- **Dirt:** any visible dirt, dust, crumb, mark, smear, or residue on the surface.
- **Standing liquid:** pooled or beaded liquid still visible on the surface.

- **Damp sheen:** a thin film of moisture with no beads and no pools. The wet pattern leaves it behind and the dry pattern takes it off.
- **Pass:** one continuous sweep of a cloth from the left side of the wipe zone to the right, parallel to the front table edge, in contact with the surface from start to finish and pressed only hard enough to keep the cloth flat under the pinch.
- **Pattern:** the full set of passes made with one cloth. Every pass runs left to right, and the passes work from the far edge toward the near edge so nothing is pushed back onto cleaned ground. Each pass overlaps the one before it by roughly half a cloth width, so no strip of the wipe zone is left untouched. The wet pattern and the dry pattern are run the same way.
- **Wet pattern:** the pattern run with the wipe, in Step 3, over the sprayed surface.
- **Dry pattern:** the pattern run with the towel, in Step 4, over the damp surface.
- **Clean and dry:** no visible dirt, no standing liquid, no damp sheen, and no streaks anywhere in the wipe zone. This is the finished state of the surface, reached at the end of Step 4.
- **Stable placement:** the object stands still after release and does not rock, lean, roll, or slide.

Steps

Step 1: Clear the items to the side

Goal: the wipe zone holds no items and every item stands on the staging strip.

Work one item per trip, front to back, so no reach passes over an item still standing.

- The **left gripper** approaches the item nearest the front of the table from above and closes on it at a point where it hangs level rather than tilted.
- Lift it clear of the surface and carry it to the staging strip. Do not drag or slide it.
- Set it down upright on the strip, clear of the items already there, and release.
- Repeat one item at a time until the wipe zone holds no items.

The right arm is the idle arm for the whole of Step 1.

Check: nothing remains inside the wipe zone, and no item on the strip is leaning, overhanging the table edge, or touching another item. If one is, settle it with the left gripper before continuing.

Step 2: Spray the surface

Goal: the wipe zone carries an even coat of liquid and nothing else on the table is wet.

- The **right gripper** closes on the spray bottle around its body, below the trigger, so the bottle hangs upright and the nozzle points away from the arms.
- Lift the bottle clear of the supply corner and carry it to the far left of the wipe zone, holding the nozzle a hand width above the surface with the trigger facing the left arm.

- The **left gripper** comes in from the left, closes on the trigger, and squeezes to fire a burst, then opens between bursts. It rests no weight on the bottle. The right gripper carries the bottle on its own the whole time.
- The **right gripper** aims each burst and may tilt the bottle so the nozzle points down at the patch being coated. Do not tilt it so far that the nozzle points level or upward, and do not tilt it while the left gripper is squeezing.
- Work in even bursts, left to right and far edge to near edge, until the coat is even with no dry patch and no pool. The right gripper moves the bottle to the next spot only after the left gripper has opened.
- Keep the nozzle inside the wipe zone for every burst. Never spray toward the staging strip, the supply corner, the bin, or a camera, and never spray into the wipe or the towel.

The **left gripper** opens off the trigger and returns to home. The **right gripper** then carries the bottle back to the supply corner, stands it upright in its start spot, and releases.

Check: the wipe zone is evenly wet, with no dry patch and no pool, the bottle stands upright in the supply corner, and the left arm is back at home with its gripper open. If an area was missed, spray it now, before the wet pattern starts, not after.

Step 3: Run the wet pattern with the wipe, then bin it

Goal: the dirt and the standing liquid are off the wipe zone and the used wipe is in the bin.

- The **right gripper** closes on the wipe with a middle pinch, carries it to the far left of the wipe zone at the far edge, and lowers it until the body of the cloth settles flat. Leave the towel where it is.
- Run the wet pattern from the far edge to the near edge. Do not lift the wipe part way through a pass, do not wipe in circles or on the diagonal, and do not regrip it by a corner.
- As soon as the last pass is done, the **right gripper** carries the wipe straight to the bin, holds it over the mouth, and opens to drop it in. The wipe never goes back to the supply corner and is never used again.

The left arm is the idle arm for the whole of Step 3.

Check: no dirt and no standing liquid in the wipe zone, no strip left untouched between passes, and the wipe in the bin with the right gripper open and empty. A damp sheen is expected here and comes off in Step 4. Anything still dirty or wet is worked with one short towel pass at the start of Step 4.

Step 4: Run the dry pattern with the towel

Goal: the wipe zone is left clean and dry, and the towel is back in the supply corner.

- The **right gripper** closes on the towel with a middle pinch, carries it to the far left of the wipe zone at the far edge, and lowers it until the body of the cloth settles flat.

- Run the dry pattern from the far edge to the near edge, the same way as Step 3, until the surface is dry to the eye.
- The **right gripper** carries the towel back to the supply corner and sets it down flat in its start spot. Never set it down inside the wipe zone and never drop it in the bin.

The left arm is the idle arm for the whole of Step 4.

Check: the wipe zone is clean and dry, with no dirt, no damp sheen, and no streaks, and no strip left untouched between passes. Work a damp patch or a streak with one short repeat pass, and cover a skipped strip with one pass. Do not restart the pattern. The items do not go back until this check passes.

Step 5: Return the items to the layout

Goal: the layout is restored and the staging strip is empty.

Work one item per trip, back to front, so no reach passes over an item already placed.

- The **left gripper** closes on the item at the back of the staging strip and lifts it clear.
- Carry it to its home position in the wipe zone and set it down in the orientation it started in, judged by eye against the table edges and the items already placed.
- Release once it is standing, then take the next item.
- Repeat one item at a time until the staging strip is empty.

Do not drag or slide an item into place. Lift it, carry it, and set it down.

The right arm is the idle arm for the whole of Step 5. It does not assist with an item.

Check: the layout is restored. If an item is off its home position, out of its orientation, leaning, or touching another item, lift it clear with the left gripper and place it again once.

Step 6: End the episode

- Confirm the wipe zone is clean and dry, the layout is restored, the staging strip is empty, the used wipe is in the bin, and the spray bottle and the towel are both back in the supply corner with the bottle upright.
- Return both arms home with grippers open, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Confirm the items are back in the layout, then spread dirt over the surface again inside the wipe zone so the next episode has something to clean.
2. Empty the bin and stand it back beside the supply corner, open at the top.
3. Stage a fresh dry wipe, spread flat in its spot in the supply corner and clear of the bottle.

4. Take the used towel away and stage a fresh dry one, spread flat in its own spot in the supply corner, clear of the bottle and of the wipe.
5. Top up the spray bottle and stand it upright in its start spot with the nozzle open.
6. Confirm the staging strip is empty and the table holds nothing but the layout items and the supplies.
7. Replace any item that is chipped, damp, or will not stand upright unaided.
8. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Item left on the surface at spray time

- **Visible cue:** the trigger is pulled in Step 2 while one or more items still stand inside the wipe zone, or spray visibly lands on an item.
- **SOP rule broken:** Step 1 (the wipe zone holds no items before Step 2 begins).
- **Coaching note:** clear every item to the strip first. Sweep the zone by eye before reaching for the bottle.

Violation: Items not carried one at a time

- **Visible cue:** in Step 1 or Step 5 an item is pushed, slid, or scraped across the table instead of being lifted clear and carried, or two or more items are moved together in one trip.
- **SOP rule broken:** Steps 1 and 5 (lift each item clear of the surface and carry it, one item per trip).
- **Coaching note:** the item leaves the table before it moves, and it travels alone. Dragging smears dirt and topples it.

Violation: Items staged badly

- **Visible cue:** at the end of Step 1 an item on the staging strip is leaning, toppled, overhanging the table edge, touching another item, or still inside the wipe zone.
- **SOP rule broken:** Step 1 (set each item down upright on the strip, clear of the items already there).
- **Coaching note:** upright and separated. A toppled item on the strip costs the return.

Violation: Spray aimed wrong

- **Visible cue:** a burst is aimed at or lands on the staging strip, the supply corner, the bin, a cloth, a camera, or the floor; the nozzle crosses outside the wipe zone while the trigger is pulled; or the right gripper tilts the bottle until the nozzle points level or upward, or tilts it while the left gripper is squeezing.
- **SOP rule broken:** Step 2 (keep the nozzle inside the wipe zone and pointed down at the patch for every burst, and move the bottle only after the left gripper has opened off the trigger).
- **Coaching note:** nozzle down inside the zone before the left gripper squeezes, left gripper open before the right gripper moves. Liquid lands on the table, never on a cloth.

Violation: Left gripper leans on the bottle

- **Visible cue:** during Step 2 the left gripper takes some of the bottle weight, pushes the bottle off the line the right gripper is holding, or stays closed on the trigger between bursts.
- **SOP rule broken:** Step 2 (the left gripper rests no weight on the bottle and opens between bursts).
- **Coaching note:** the right gripper carries the bottle on its own. The left gripper squeezes and lets go.

Violation: Bottle not returned upright

- **Visible cue:** at the end of Step 2 the bottle lies on its side, is left inside the wipe zone, is still in the gripper during Step 3, or is dropped rather than set down.
- **SOP rule broken:** Step 2 (stand the bottle upright in its start spot in the supply corner and release).
- **Coaching note:** stand the bottle back where it started before you pick up the wipe.

Violation: Cloth not pinched at the middle

- **Visible cue:** in Step 3 or Step 4 the cloth is held by a corner or an edge, balled up in the gripper, or dragged edge on, so only a narrow line of cloth touches the surface, or it is regripped by a corner part way through the pattern.
- **SOP rule broken:** Steps 3 and 4 (close on the cloth at its middle so its body settles flat on the surface).

- **Coaching note:** pinch the middle and let the cloth hang even. A corner grip cleans a line, not a strip.

Violation: Pattern not followed

- **Visible cue:** in Step 3 or Step 4 passes run right to left, front to back, on the diagonal, or in circles; a strip of the wipe zone is left untouched between two passes or the passes sit a full cloth width or more apart; or the cloth lifts off part way through a pass instead of running as one continuous contact from left to right.
- **SOP rule broken:** Steps 3 and 4 (every pass left to right and parallel to the front edge, in contact from start to finish, working far edge to near edge, each pass overlapping the one before it by roughly half a cloth width).
- **Coaching note:** hold contact for the full width, lift at the end, return to the left, step in by about half a width. Gaps between passes are the most common cause of a failed wipe.

Violation: Wipe and towel used out of role

- **Visible cue:** the towel is taken out of the supply corner for the wet pattern in Step 3, the wipe is used for the dry pattern in Step 4, both cloths are carried at once, or the two are used in the wrong order.
- **SOP rule broken:** Steps 3 and 4 (the wipe runs the wet pattern, the towel runs the dry pattern).
- **Coaching note:** wipe first on the wet surface, towel second on the damp one. Check which cloth is in the gripper before the first pass.

Violation: Cloth not parked where it belongs

- **Visible cue:** the used wipe goes back to the supply corner, is left on the table, is kept in the gripper into Step 4, or misses the bin and lands on the floor; the towel is dropped in the bin, left in the wipe zone, or draped over an item instead of being set down flat in its spot in the supply corner.
- **SOP rule broken:** Steps 3 and 4 (drop the wipe into the bin as soon as the wet pattern ends, and set the towel down in its start spot in the supply corner).
- **Coaching note:** the wipe goes in the bin, the towel parks in the corner. A used cloth back on the clean surface undoes the whole wipe.

Violation: Dirt, standing liquid, or streaks left in the wipe zone

- **Visible cue:** at the end of Step 3 visible dirt or pooled liquid remains in the wipe zone, or at the end of Step 4 a damp patch, a streak, or dirt remains, and the episode moves on without a repeat pass over the spot.
- **SOP rule broken:** Steps 3 and 4 (the zone is left clean and dry, with one short repeat pass over anything that remains).

- **Coaching note:** check the zone at the end of each pattern, then take the one repeat pass. The items go back on a clean dry surface.

Violation: Dry pattern skipped or cut short

- **Visible cue:** Step 4 is skipped entirely and the items go back on a damp surface, the towel is never taken out of the supply corner, or the dry pattern stops before its last pass reaches the near edge of the wipe zone.
- **SOP rule broken:** Step 4 (run the dry pattern with the towel over the whole wipe zone before the items go back).
- **Coaching note:** the wet pattern is only half the job. The towel covers the same ground before anything is returned.

Violation: Items returned to a wet surface

- **Visible cue:** an item is set back into the wipe zone in Step 5 while standing liquid or a damp sheen is still visible, or before Step 4 is finished.
- **SOP rule broken:** Step 5 (return only after the dry pattern is complete and the surface shows no standing liquid and no damp sheen).
- **Coaching note:** finish and check the dry pattern first. A wet surface marks the items.

Violation: Phase order broken

- **Visible cue:** the phases run out of order or a finished phase is revisited: spraying after the wet pattern has started, wiping before the spray, drying before the wipe is binned, returning items before the dry pattern is finished, or clearing an item again after Step 1 is done.
- **SOP rule broken:** Steps 1 to 5 (clear, spray, wet pattern, dry pattern, return, each finished and checked before the next begins).
- **Coaching note:** finish a phase, check it, move on. Never go back to one already done.

Violation: Wrong arm used for an action

- **Visible cue:** the left gripper touches the wipe or the towel, or touches the bottle anywhere other than the trigger in Step 2, or takes the bottle off the supply corner or carries it; the right gripper moves an item or presses the trigger; the bottle passes from one gripper to the other; or a step that names a gripper is performed with the other one.
- **SOP rule broken:** Steps 1 to 5 (each arm works only the actions assigned to its gripper).
- **Coaching note:** left moves items and presses the trigger, right handles tools. Nothing else swaps and nothing changes hands.

Violation: Grip or placement fumbled

- **Visible cue:** the gripper closes on an item, the bottle, the trigger, or a cloth, finds the grip off, opens, and closes again more than twice before lifting or firing; or more than about two small corrections are made to settle an item, the bottle, or a pass.

- **SOP rule broken:** Steps 1 to 5 (approach from above, grasp securely on the first close, and settle in one or two small corrections).
- **Coaching note:** the approach angle is off and is forcing correction. Line the gripper up before closing so the first grip catches and each placement lands close to target.

Violation: Item dropped, collided, or knocked over

- **Visible cue:** an item, the bottle, or a cloth falls or is struck; an item topples on the strip or in the layout; an arm knocks the bottle over in the supply corner; or an arm knocks the bin over or drags it out of place.
- **SOP rule broken:** Steps 1 to 5 (carry every object clear of what it passes over and release only after a stable placement).
- **Coaching note:** check the path and the landing spot before you move.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with dirt or damp in the wipe zone, an item on the staging strip, the used wipe anywhere but the bin, the towel off its spot in the supply corner, the bottle off its spot or on its side, an arm close to home but not at it, or a gripper not fully open.
- **SOP rule broken:** Step 6 (confirm the clean dry zone, the restored layout, the empty strip, the wipe in the bin, and the bottle and towel back in the supply corner; return both arms home with grippers open; then stop recording).
- **Coaching note:** confirm, home, stop. Complete the home motion before you end the recording.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Defective spray bottle:** a nozzle that is blocked or leaking, a trigger that will not fire under a gripper close, or a bottle that runs empty part way through. Replace or refill before the next episode.
- **Defective wipe:** a wipe that is torn, shedding, already wet at episode start, or so soaked part way through the wet pattern that it stops lifting liquid. Replace before the next episode.
- **Defective towel:** a towel that is torn, shedding, already damp at episode start, or so soaked part way through the dry pattern that it stops drying. Replace before the next episode.
- **Damaged surface:** a stain, scratch, or mark in the surface itself that does not lift under a correct wet pattern. Move the wipe zone or replace the surface before the next episode.

- **Defective item:** an item that will not stand upright unaided or slips out of a correct grip. Replace before the next episode.
- **The staging strip, the supply corner, or the bin shifts out of its zone** during the episode.

Annotation subtasks (from SOP)

1. Move one item from the wipe zone to the staging strip
2. Take the spray bottle from the supply corner
3. Bring the left gripper onto the trigger of the held bottle
4. Fire one burst onto the wipe zone
5. Return the left gripper to home off the trigger
6. Return the spray bottle to the supply corner
7. Take the wipe from the supply corner with a middle pinch
8. Run one pass with the wipe
9. Drop the used wipe into the bin
10. Take the towel from the supply corner with a middle pinch
11. Run one pass with the towel
12. Return the towel to the supply corner
13. Move one item from the staging strip to its home position
14. Return both arms home and end the episode